

Lab Problem 10

Objective: Compare Linear and RBF SVM models using accuracy and decision boundary visualization.

Dataset:

- Two-moons synthetic dataset (200 samples, noise = 0.15)
- Use a fixed random state for reproducibility

Tasks:

1. Data Preparation

- Generate dataset
- Train-test split (80:20)
- Apply feature scaling

2. Linear SVM

- Train Linear SVM ($C = 1$)
- Evaluate test accuracy

3. RBF SVM Models

Train three RBF kernel SVM models with:

- Model A: $\gamma = 0.1$, $C = 1$
- Model B: $\gamma = 1$, $C = 1$
- Model C: $\gamma = 5$, $C = 10$

4. Evaluation

- Evaluate all models on the test dataset
- Report accuracy for each model
- Present results in a clear, organized format

Visualization:

1. Decision Boundaries

Plot decision boundaries for:

- Linear SVM
- Model A, B, and C

2. Support Vectors

- Identify and plot support vectors for RBF models

3. Accuracy Comparison

- Create a bar chart comparing model accuracies